

# Hands-On-Learning: Evidence Handling and Hashing Integrity Verification

# Overview

Learning Objective Alignment: LO2 – Evidence collection. This lab provides hands-on practice with the core skills of evidence documentation (chain of custody) and integrity verification (forensic hashing).

# Scenario

As a junior forensic investigator, you have been dispatched to a small business, "Innovate LLC," to collect a laptop belonging to a sales employee, "Jane Smith," who is suspected of emailing a confidential client list to a competitor. You have the proper legal authority to seize the device. When you arrive, you find the laptop is on and logged in, with an email program open on the screen. You will photograph the scene, document the seizure, transport the evidence, and then use a standard forensic tool to verify the integrity of a pre-made forensic image. You will also be asked to identify a potential encrypted volume within the image.

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# Pre-Lab Requirements

## Tools:

A computer with Windows, macOS, or Linux.

An internet connection.

A pre-downloaded copy of a forensic hashing utility. We will use FTK Imager from AccessData, which is a free and widely used tool. (A download link is provided in the Additional Resources section).

A sample forensic image file to test. (A download link for a small sample disk image is provided).

**Environment:** A working directory on your computer to store the lab files.

**Prerequisites:** You must have completed all nine videos for Module 2:

Lesson 1: Digital Evidence Sources, Crime Scene Procedures, Volatile Evidence Handling.

Lesson 2: Chain of Custody Principles, Evidence Packaging and Transport, Storage and Access Controls.

Lesson 3: Forensic Imaging Techniques, Hash Verification and Integrity, Working with Encrypted Data.

# Step-by-Step Walkthrough

# Phase 1 – On-Scene Procedures (Simulation)

Step 1: Create a new document using a text editor or word processor. Title it "Case Notes - Case #2025-001."

Step 2: Your first action on scene is to handle the live system. In your notes, describe the single most important piece of volatile data visible (the open email program) and state that you would photograph the screen before taking any other action. This fulfills the requirement to document the scene.

# Phase 2 – Evidence Documentation (Simulation)

Step 1: Create a new document using a text editor or word processor. Title it "Chain of Custody - Case #2025-001."

Step 2: Create a simulated Chain of Custody form in this document. It must include fields for: Case Number, Item Number, Description of Item, Date/Time of Seizure, Location of Seizure, Seized By.

Step 3: Fill out the form with the following details:

Case Number: 2025-001

Item Number: 2025-001-A

Description: One (1) silver Dell Latitude Laptop, Model 7420, Serial #ABC123XYZ. Minor scratch on top cover, company asset tag #55-1234 attached.

Date/Time of Seizure: [Use today's date and the current time]

Location of Seizure: Innovate LLC, 123 Business Rd, Anytown, USA.  
Seized By: [Your Name]

Step 4: Below this initial seizure block, create a transfer log with columns: Date/Time, Released By, Received By, and Purpose. Create an entry showing you (Released By) transferring the laptop to the Lab Evidence Locker (Received By) for Secure Storage.

Expected Result: A completed, professional-looking chain of custody form documenting the initial seizure and transfer of the laptop.

# Phase 3 – Forensic Image Analysis

Step 1: Install FTK Imager on your computer.

Step 2: Download the sample disk image file (sample\_image.E01) from the provided link and save it to your working directory.

Step 3: Along with the image, you are given the original SHA-256 hash value that was "calculated from the source drive" at the time of acquisition.

Original SHA-256 Hash:

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 (Note: This is the hash of an empty file, which we use here for simplicity).

Step 4: Launch FTK Imager. Go to File -> Verify Image/Disk. Browse to and select the sample\_image.E01 file you downloaded. Compare the computed SHA-256 hash with the original hash provided.

Step 5: Now, add the sample\_image.E01 as evidence. Go to File -> Add Evidence Item. Select the image file. In the Evidence Tree pane on the left, expand the image and look for a file named encrypted\_archive.zip. This is a common indicator of an encrypted container.

Step 6: In your case notes document, add a section titled "Image Analysis Findings." Note that the hash verification was successful and that you identified a potential encrypted file named encrypted\_archive.zip that requires further investigation and possible password cracking.

Expected Result: The hash values should match, and your notes should reflect the discovery of the encrypted file.

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Original SHA-256

Hash:e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 (Note: This is the hash of an empty file, which we use here for simplicity).

Step 4: Launch FTK Imager. Go to File -> Verify Image/Disk. Browse to and select the sample\_image.E01 file you downloaded.

Step 5: The tool will process the file and calculate its hash values. It will display the computed MD5 and SHA-1 hashes.

Step 6: Compare the SHA-256 hash value computed by FTK Imager with the Original SHA-256 Hash provided in Step 3.

Expected Result: The hash values should match exactly. This confirms the integrity of the forensic image.

# Deliverables

**1** Case Notes Document: Your completed Case Notes and Chain of Custody document from all phases (saved as [YourLastName]\_M2\_Lab\_Notes.pdf).

**2** Screenshot of Verification: A screenshot of the FTK Imager window showing the computed hash values for the sample image. The screenshot must clearly show the filename and the hash results. (Saved as [YourLastName]\_M2\_Lab\_Hash.png).

# Assessment Criteria

Chain of Custody Form (50%): Is the form complete, professional, and does it contain all the required information from the scenario?

Hashing Verification (50%): Does the screenshot clearly show that the student successfully used the forensic tool to calculate the hash of the provided image?

# Reflection Questions

- 1 Why was it important to photograph the live screen before unplugging the laptop?
- 2 What would it mean for the case if the hashes did not match? What are your ethical and procedural obligations as an examiner at that point?
- 3 Now that you've identified an encrypted file, what are the potential legal and technical challenges you face in accessing its contents?

# Additional Resources

- FTK Imager Download: [Link to the official FTK Imager download page]
- Sample Disk Image: [Link to a downloadable .E01 sample file]

# Instructions for Documenting and Sharing The Project for Peer Review

## Document the Project



Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



## Share the Project

Document: Save the report in PDF format.

**Upload:** Share in the “Peer Review” area of the course shell with a brief description.



## Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.